

Parallel Molecular Dynamics

Aiichiro Nakano

*Department of Computer Science
Department of Physics & Astronomy
Department of Quantitative & Computational Biology
University of Southern California*

Email: anakano@usc.edu

Objective: Operationally understand spatial decomposition (who does what) & message passing using a real-world application (pmd.c)

<https://aiichironakano.github.io/cs596/src/pmd>
<https://github.com/KenichiNomura/binary-LJ-pmd>

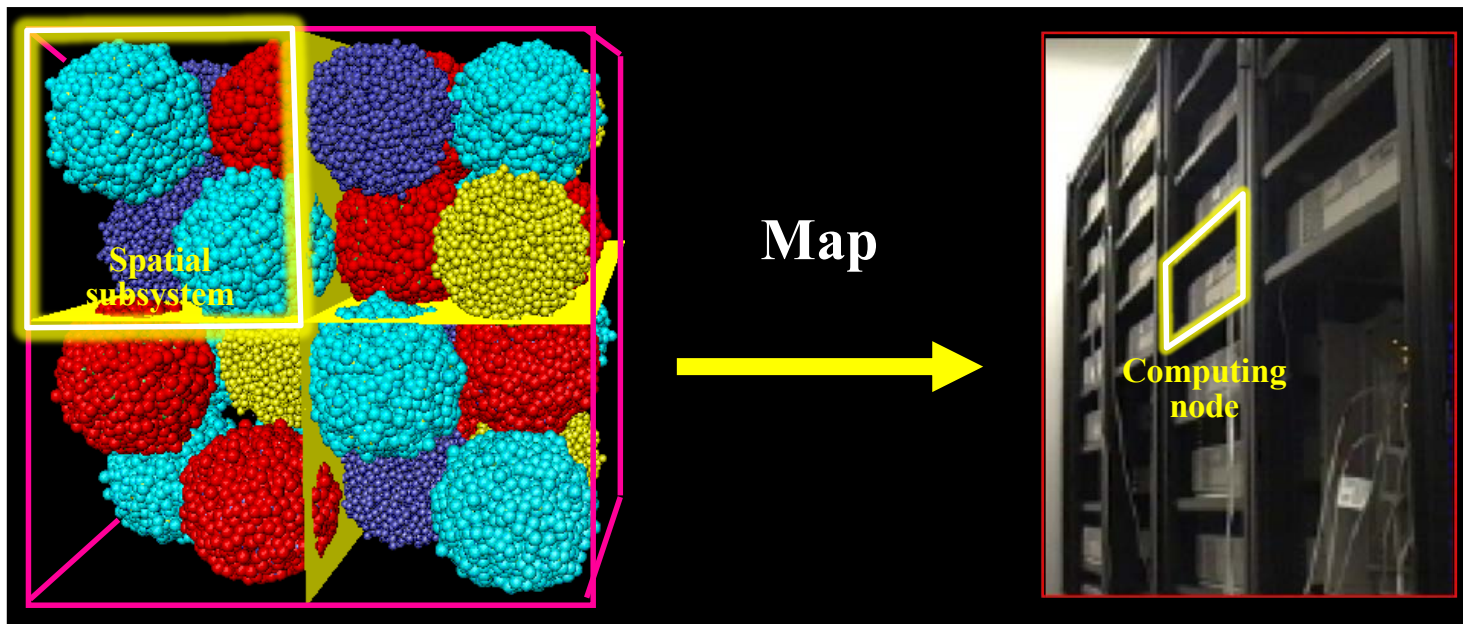


Parallel Molecular Dynamics

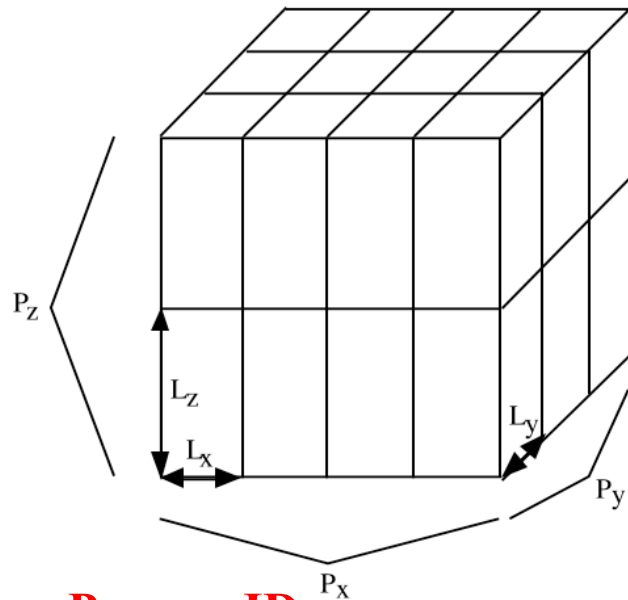
Spatial decomposition (short-ranged):

Will learn other decomposition schemes later:
<https://aiichironakano.github.io/cs596/NT.pdf>

1. Divide the physical space into subspaces of equal volume
2. Assign each subspace to a computing node (more generally, to a process) in a parallel computer
or MPI rank
3. Each node computes forces on the atoms in its subspace & updates their positions & velocities
Who does what



Spatial Decomposition



- Process ID**

Vector

$$p_x = p / (P_y P_z)$$

$$p_y = (p / P_z) \bmod P_y$$

$$p_z = p \bmod P_z$$

Scalar

$$p = p_x \times P_y \times P_z + p_y \times P_z + p_z \quad \text{Rank}$$

Which 3D
subspace?

*Map a spatial
subsystem to
a process!*

$P_z - 1$

5	11	17	23
3	9	15	21
1	7	13	19

cf. MD lecture
slides 29 & 30

$P_y - 1$

4	10	16	22
2	8	14	20
0	6	12	18

$P_x - 1$

0 1 $P_x - 1$

P_z P_y P_x

$$nproc = vproc[0] \times vproc[1] \times vproc[2]$$

In pmd.h

int vproc[3] = {1,1,2}, nproc = 2;

In pmd.c

MPI_Comm_rank(MPI_COMM_WORLD, &sid);
vid[0] = sid / (vproc[1] * vproc[2]);
vid[1] = (sid / vproc[2]) % vproc[1];
vid[2] = sid % vproc[2];

Neighbor Process ID

$$p'_\alpha(\kappa) = [p_\alpha + \delta_\alpha(\kappa) + P_\alpha] \bmod P_\alpha \quad (\kappa = 0, \dots, 5; \alpha = x, y, z) \text{ neighbor's vector process ID}$$

$$p'(\kappa) = p'_x(\kappa) \times P_y P_z + p'_y(\kappa) \times P_z + p'_z(\kappa) \text{ neighbor's rank}$$

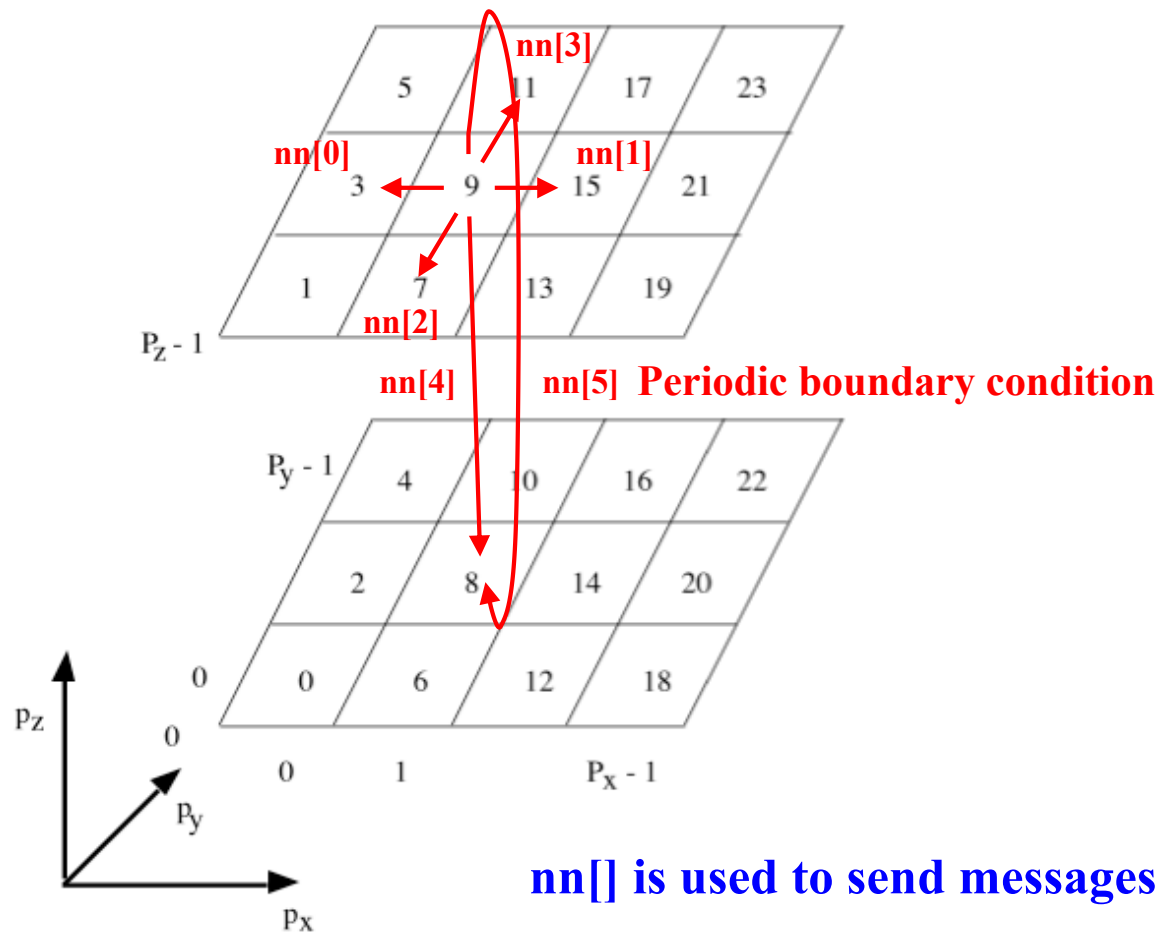
Neighbor ID, κ	$\vec{\delta} = (\delta_x, \delta_y, \delta_z)$	$\vec{\Delta} = (\Delta_x, \Delta_y, \Delta_z)$
0 (east)	$(-1, 0, 0)$	$(-L_x, 0, 0)$
1 (west)	$(1, 0, 0)$	$(L_x, 0, 0)$
2 (north)	$(0, -1, 0)$	$(0, -L_y, 0)$
3 (south)	$(0, 1, 0)$	$(0, L_y, 0)$
4 (up)	$(0, 0, -1)$	$(0, 0, -L_z)$
5 (down)	$(0, 0, 1)$	$(0, 0, L_z)$

- L_x , L_y & L_z are the box lengths *per process* in the x, y & z directions
- Atom coordinates are in the range $[0, L_\alpha]$ ($\alpha = x, y, z$) in each process

In pmd.c

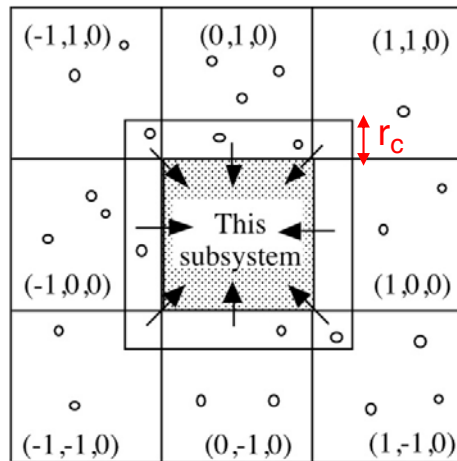
[illegible]

Neighbor Process ID Example

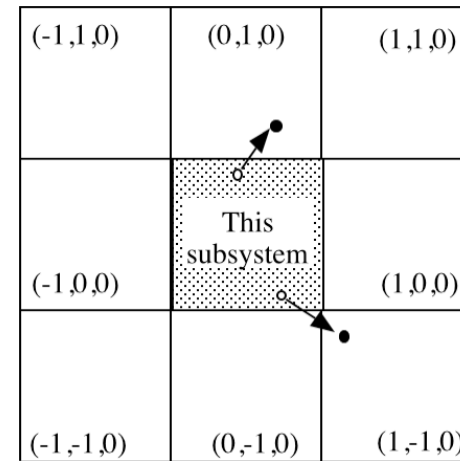


Parallel MD Concepts

Atom caching

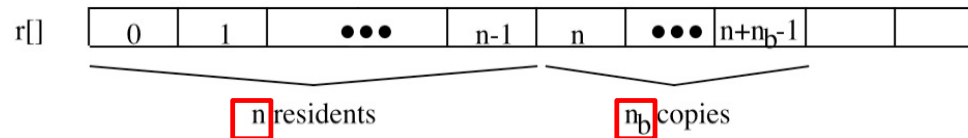


Atom migration



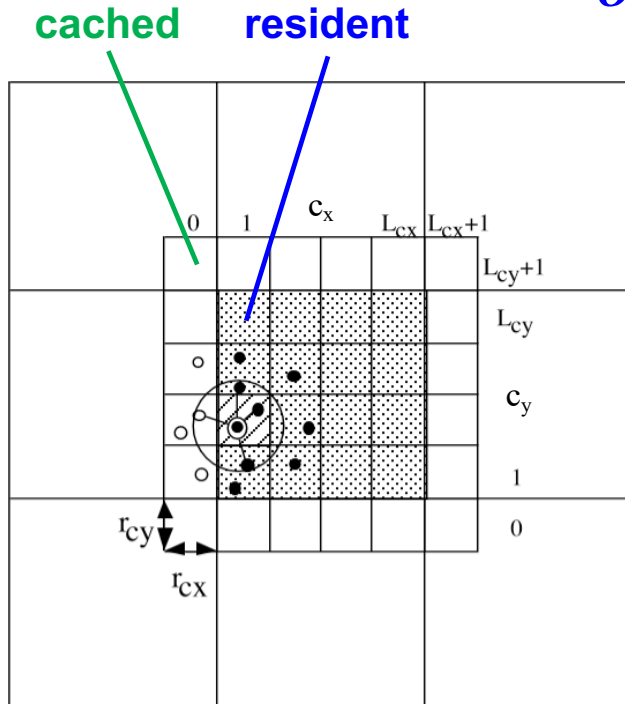
1. First half kick to obtain $v_i(t+Dt/2)$
2. Update atomic coordinates to obtain $r_i(t+Dt)$
3. `atom_move()`: Migrate the moved-out atoms to the neighbor processes
4. `atom_copy()`: Copy the surface atoms within distance r_c from the neighbors
5. `compute_accel()`: Compute new accelerations, $a_i(t+Dt)$, including the contributions from the cached atoms
6. Second half kick to obtain $v_i(t+Dt)$

Data structure



Linked-List Cell Method

Search for pairs only within the nearest neighbor cells:

$$O(N^2) \rightarrow O(N)$$


- **Cell size**

$$L_{c\alpha} = \lfloor L_{\alpha} / r_c \rfloor$$

$$r_{c\alpha} = L_{\alpha} / L_{c\alpha} \quad (\alpha = x, y, z)$$
- **Cell index**

$$c = c_x(L_{cy}+2)(L_{cz}+2) + c_y(L_{cz}+2) + c_z$$

$$c_x = c / [(L_{cy}+2)(L_{cz}+2)]$$

$$c_y = [c / (L_{cz}+2)] \bmod (L_{cy}+2)$$

$$c_z = c \bmod (L_{cz}+2)$$
- **Atom $\rightarrow$ cell mapping**

$$c_{\alpha} = \lfloor (r_{\alpha} + r_{c\alpha}) / r_{c\alpha} \rfloor \quad (\alpha = x, y, z)$$

Only change from serial lmd.c in green: Augmented cells to include cached atoms

List Construction Algorithm

```

/* Reset the headers, head */
for (c=0; c<lcxyz2; c++) head[c] = EMPTY;
/* Scan atoms to construct headers, head, & linked lists, lscl */
for (i=0; i<n+nb; i++) { Consider  $n_b$  cached atoms
  /* Vector cell index to which this atom belongs */
  for (a=0; a<3; a++) mc[a] = (r[i][a]+rc[a])/rc[a]; Position offset by one cell
  /* Translate the vector cell index, mc, to a scalar cell index */
  c = mc[0]*lcyz2+mc[1]*lc2[2]+mc[2];
  /* Link to the previous occupant (or EMPTY if you're the 1st) */
  lscl[i] = head[c];
  /* The last one goes to the header */
  head[c] = i;
}

```

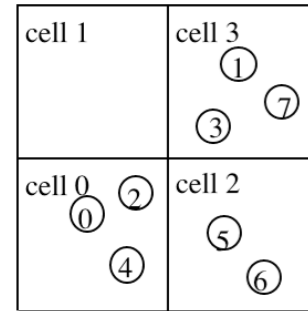
In the above:

$lcyz2 = lc2[1]*lc2[2]$

where

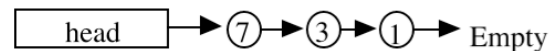
$lc2[a] = lc[a]+2$ ($a = 0,1,2$)

$lcxyz2 = lcyz2*lc2[0]$



	0	1	2	3															
head	4	E	6	7															
	0	1	2	3	4	5	6	7											
lscl	E	E	0	1	2	E	5	3											

Change from serial lmd.c in green



Interaction Computation

```

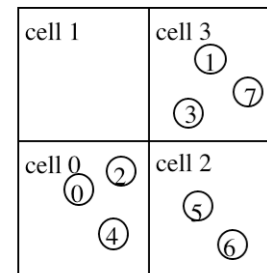
/* Scan inner cells (resident) */
for (mc[0]=1; mc[0]<=lc[0]; (mc[0]))++
for (mc[1]=1; mc[1]<=lc[1]; (mc[1]))++
for (mc[2]=1; mc[2]<=lc[2]; (mc[2]))++ {
    /* Calculate a scalar cell index */
    c = mc[0]*lcyz2+mc[1]*lc2[2]+mc[2];
    /* Scan the neighbor cells (including itself) of cell c (resident + cached) */
    for (mc1[0]=mc[0]-1; mc1[0]<=mc[0]+1; (mc1[0]))++
    for (mc1[1]=mc[1]-1; mc1[1]<=mc[1]+1; (mc1[1]))++
    for (mc1[2]=mc[2]-1; mc1[2]<=mc[2]+1; (mc1[2]))++ {
        /* Calculate the scalar cell index of the neighbor cell */
        c1 = mc1[0]*lcyz2+mc1[1]*lc2[2]+mc1[2];
        /* Scan atom i in cell c */
        i = head[c];
        while (i != EMPTY) {
            /* Scan atom j in cell c1 */
            j = head[c1];
            while (j != EMPTY) {
                ...
                if (i<j && rij<rc2) Process pair (i, j)
                ...
                j = lscl[j];
            }
            i = lscl[i];
        }
    }
}

```

Who does what: Each rank computes forces on the **resident atoms in its subspace & updates their positions & velocities**

Resident atoms may interact with **cached atoms (cf. slide 7)**

Change from serial lmd.c in green



	0	1	2	3				
head	4	E	6	7				
	0	1	2	3	4	5	6	7
lscl	E	E	0	1	2	E	5	3



Parallel Interaction Computation

SPMD: Who does what?

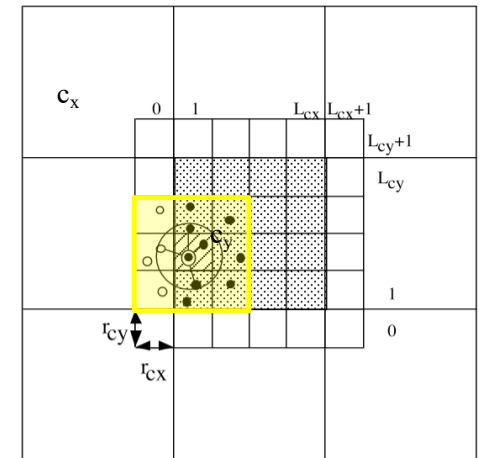
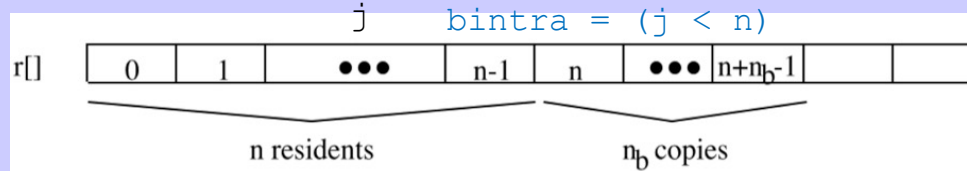
Each process computes:

1. The forces on its resident atoms
2. The potential energy between resident pairs & 1/2 of that between resident-cached pairs

```

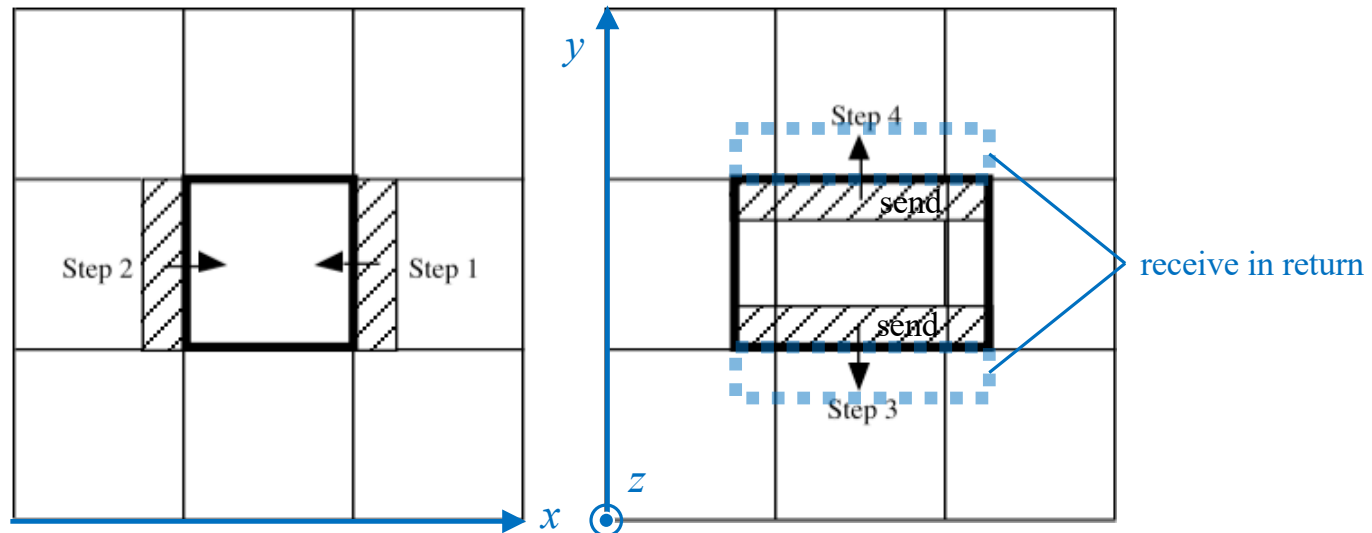
for resident cells, c {
  for neighbor (resident or cached) cells, c1 {
    scan atom i in cell c using c's linked list {
      scan atom j in cell c1 using c1's linked list {
        ...
        if (i < j && rij < rc2) {
          compute pair force aij & potential u(rij)
          bintra = j < n; // j is resident?
          ai += aij; if (bintra) aj -= aij;
          if (bintra) lpe += u(rij); else lpe += u(rij)/2;
        }
      }
    }
  }
}

MPI_Allreduce(&lpe, &potEnergy, ..., MPI_SUM, ...);
  
```



Atom Caching: `atom_copy()`

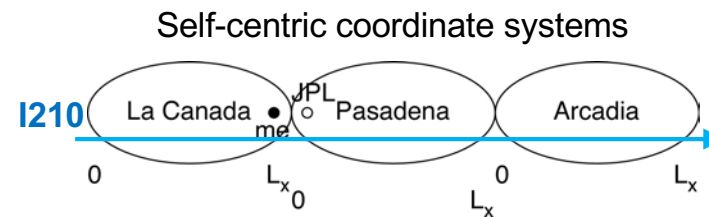
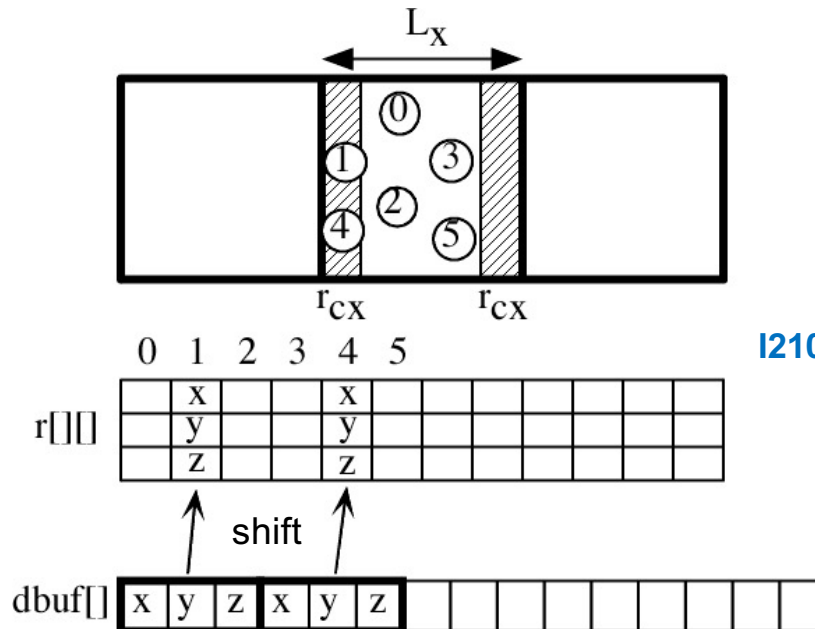
Caching from 26 neighbors in 6 steps (by forwarding)



```

Reset the number of received cache atoms, nbnew = 0
for x, y, and z directions
  Make boundary-atom lists, lsb, for lower and higher directions including both
  resident, n, and cache, nbnew, atoms (within  $r_c$  from boundary)
  for lower and higher directions
    Send/receive boundary-atom coordinates to/from the neighbor
    Increment nbnew;
  endfor
endfor
nb = nbnew
  
```

Implementing Atom Caching



Copying condition

```

bbd(ri[], ku) {
    kd = ku / 2 (= 0|1|2) x|y|z
    kdd = ku % 2 (= 0|1) lower|higher
    if (kdd == 0)
        return ri[kd] < RCUT
    else
        return al[kd] - RCUT < ri[kd]
}

```

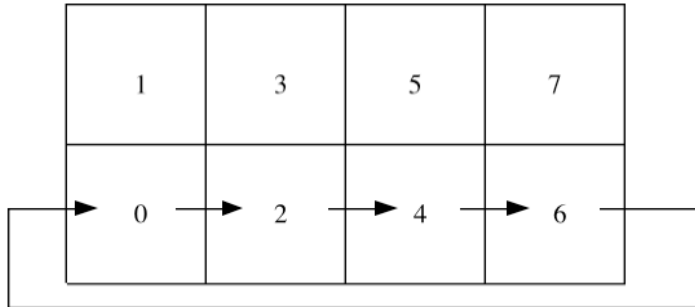
3 phases of message passing

1. Message buffering: $dbuf \leftarrow r - sv$ (shift), gather [See slide 4 for sv\[\]](#)
2. Message passing: $dbuf \leftarrow dbuf$
Send dbuf
Receive dbuf
3. Message storing: $r \leftarrow dbuf$, append after the residents

See `atom_copy()` in `pmd.c`

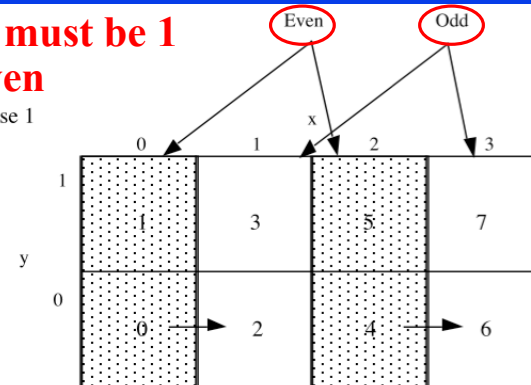
Deadlock Avoidance

Cyclic dependence

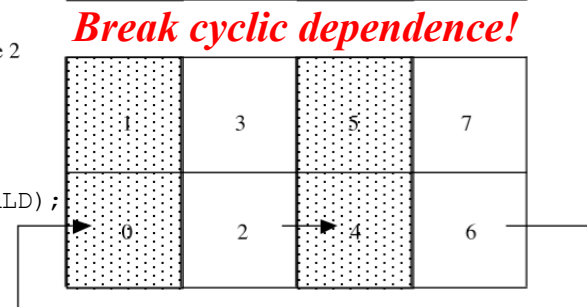


vproc[0|1|2] must be 1 or even

Phase 1



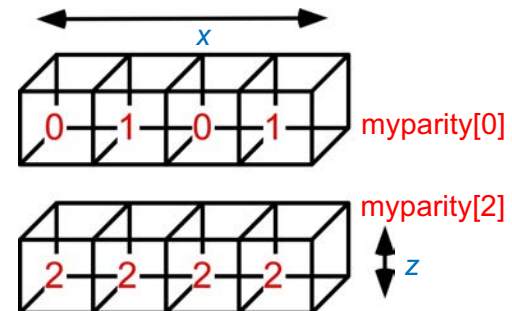
Phase 2



Break cyclic dependence!

3-phase (deadlock-free) message passing

1. Message buffering: $\text{dbuf} \leftarrow r$, gather
2. Message passing: $\text{dbufr} \leftarrow \text{dbuf}$
 /* Even node: send & recv, if not empty */
 if ($\text{myparity}[\text{kd}] == 0$) {
 $\text{MPI_Send}(\text{dbuf}, 3*\text{nsd}, \text{MPI_DOUBLE}, \text{inode}, 120, \text{MPI_COMM_WORLD});$
 $\text{MPI_Recv}(\text{dbufr}, 3*\text{nrc}, \text{MPI_DOUBLE}, \text{MPI_ANY_SOURCE}, 120,$
 $\text{MPI_COMM_WORLD}, \&\text{status});$
 }
 /* Odd node: recv & send, if not empty */
 else if ($\text{myparity}[\text{kd}] == 1$) {
 $\text{MPI_Recv}(\text{dbufr}, 3*\text{nrc}, \text{MPI_DOUBLE}, \text{MPI_ANY_SOURCE}, 120,$
 $\text{MPI_COMM_WORLD}, \&\text{status});$
 $\text{MPI_Send}(\text{dbuf}, 3*\text{nsd}, \text{MPI_DOUBLE}, \text{inode}, 120, \text{MPI_COMM_WORLD});$
 }
 /* Single layer: Exchange information with myself */
 else $\text{myparity}[\text{kd}] = 2$ if $\text{vproc}[\text{kd}] = 1$
 for ($i=0; i<3*\text{nrc}; i++$) $\text{dbufr}[i] = \text{dbuf}[i];$
3. Message storing: $r \leftarrow \text{dbufr}$, append



ANL IBM SP1 User's Guide ('94)

11. Q: My parallel program runs on other parallel machines but seems to deadlock on the SP-1 when using EUI, EUI-H, or Chameleon.

A: The following parallel program can deadlock on *any* system when the size of the message being sent is large enough:

```
send( to=partner, data, len, tag )
recv( from=partner, data, maxlen, tag )
```

where these are blocking send's and receives (`mp_bsend` in EUI/EUI-H and `PIbsend` in Chameleon). For many systems, deadlock does not occur until the message is very long (often 128 KBytes or more). For EUI, the size is (roughly) 128 bytes (*not* KBytes) and for EUI-H, the size is (again roughly) 4 KBytes. The limit for Chameleon is the same as the underlying transport layer (i.e., the EUI or EUI-H limits).

To fix this you have several choices:

- pmd.c** • Reorder your send and receive calls so that they are pair up. For example, if there are always an even number of processors, you could use

```
if (myid is even) {
    send( to=partner, data, len, tag )
    recv( from=partner, data, maxlen, tag )
}
else {
    recv( from=partner, data, maxlen, tag )
    send( to=partner, data, len, tag )
}
```

- assignment** • Use non-blocking sends and receives instead



```
MPI_Irecv() ;
MPI_Send() ;
MPI_Wait() ;
```

Digress: Polyacetylene & Peierls Distortion

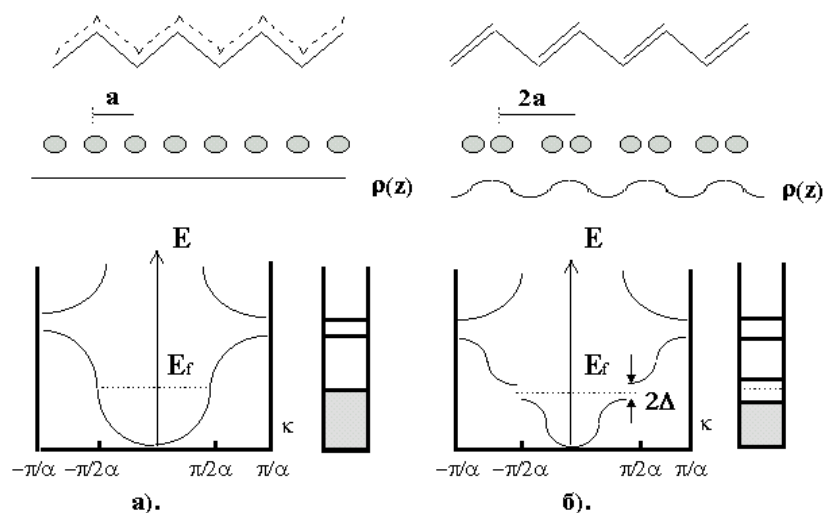
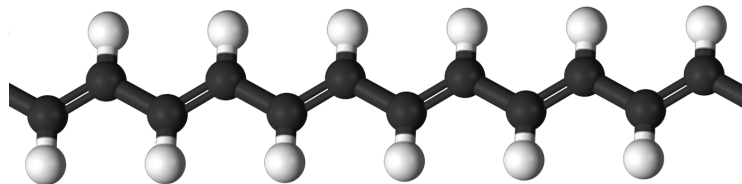


Fig 1. Electron dispersion and a band pattern of one-dimensional molecular system:
a). metallic and b). insulator state, ($\rho(z)$ —a electronic density, a —a lattice period).



Alan J. Heeger
Prize share: 1/3

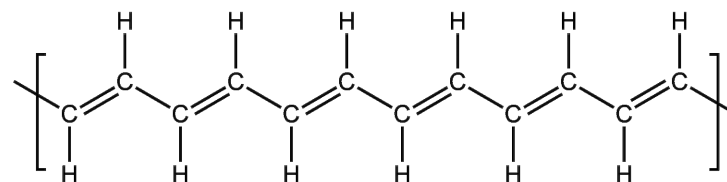


Alan G. MacDiarmid
Prize share: 1/3



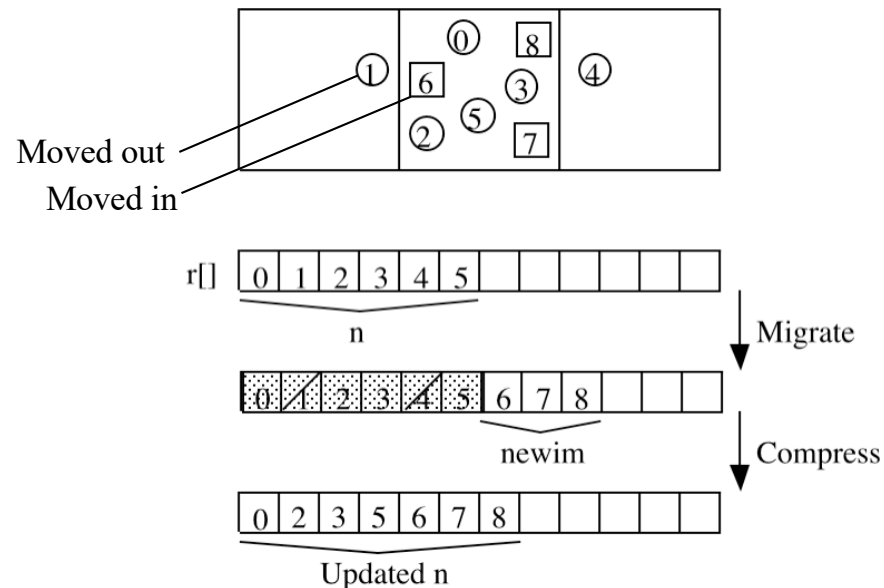
Hideki Shirakawa
Prize share: 1/3

The Nobel Prize in Chemistry 2000 was awarded jointly to Alan J. Heeger, Alan G. MacDiarmid and Hideki Shirakawa "for the discovery and development of conductive polymers".



Nature's spontaneous even-odd
symmetry breaking

Atom Migration: atom_move()



Reset the number of received new immigrants, `newim = 0`

for x, y, and z directions

Make moving-atom lists, `mvque`, for lower and higher directions including both resident, `n`, and immigrant, `newim`, atoms but excluding those already moved out for lower and higher directions

Send/receive moving-atom coordinates to/from the neighbor

(When moving, `r[][0] ← MOVED_OUT = -1010`)

Increment `newim`

endfor

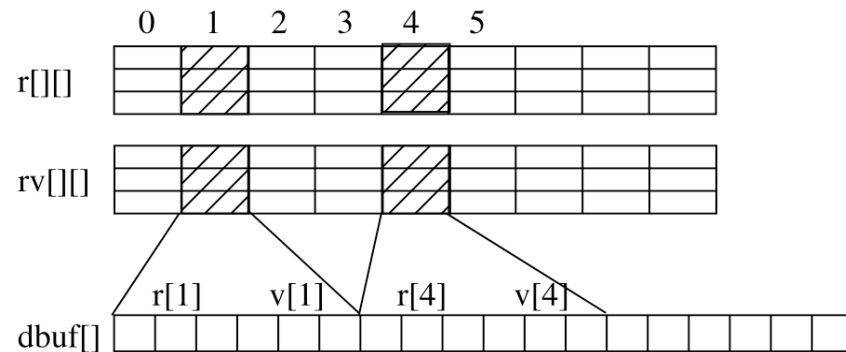
endfor

Compress the `r` array to eliminate the moved-out atoms

Implementing Atom Migration

Moving condition

```
bmv(ri[], ku) {  
    kd = ku / 2 (= 0|1|2) x|y|z  
    kdd = ku % 2 (= 0|1) lower|higher  
    if (kdd == 0)  
        return ri[kd] < 0.0  
    else  
        return al[kd] < ri[kd]  
}
```



3 phases of message passing

1. Message buffering: `dbuf` $\leftarrow$ `r`-sv (shift) & `rv`, gather
Mark `MOVED_OUT` in `r`
2. Message passing: `dbuf` $\leftarrow$ `dbuf`
Send `dbuf`
Receive `dbuf`
3. Message storing: `r` & `rv` $\leftarrow$ `dbuf`, append after the residents

See `atom_move()` in `pmd.c`

Bottom Line: Parallel MD

Parallel computing:

Specifies who does what — decomposition

Parallel molecular dynamics (spatial decomposition):

Who does what = each processor computes forces on only resident atoms in the subspace assigned to it & updates their positions & velocities

Scalability Metrics for Parallel Molecular Dynamics

Aiichiro Nakano

*Department of Computer Science
Department of Physics & Astronomy
Department of Quantitative & Computational Biology
University of Southern California*

Email: anakano@usc.edu

Objective: Consolidate your understanding of scalability analysis (*e.g.*, fixed-problem vs. isogranular scaling) using a real-world example of pmd.c

solve the same problem faster

solve a larger problem within the same time



Recap: Parallel Efficiency

Parallel computing = solving a big problem (W) in a short time (T) using many processors (P)

- **Execution time:** $T(W,P)$
 W : Workload
 P : Number of processors

- **Speed:** $S(W,P) = \frac{W}{T(W,P)}$

- **Speedup:** $S_P = \frac{S(W_P,P)}{S(W_1,1)} = \frac{W_P T(W_1,1)}{W_1 T(W_P,P)}$

- **Efficiency:** $E_P = \frac{S_P}{P} = \frac{W_P T(W_1,1)}{P W_1 T(W_P,P)}$

How to scale W_P with P ?

Fixed Problem-Size (Strong) Scaling

Solve the same problem faster using more processors

$W_P = W$ —constant (strong scaling)

$$S_P = \frac{T(W,1)}{T(W,P)} \leq P$$

• **Speedup:** $S_P = \frac{W_P T(W_1,1)}{W_1 T(W_P,P)} = \frac{T(W,1)}{T(W,P)}$ $S_P = \frac{S(W_P,P)}{S(W_1,1)} = \frac{W_P T(W_1,1)}{W_1 T(W_P,P)}$

• **Efficiency:** $E_P = \frac{T(W,1)}{P T(W,P)}$ $E_P = \frac{S_P}{P} = \frac{W_P T(W_1,1)}{P W_1 T(W_P,P)}$

- **Amdahl's law:** f (= sequential fraction of the workload) limits the asymptotic speedup

$$T(W,P) = fT(W,1) + \frac{(1-f)T(W,1)}{P}$$
$$\therefore S_P = \frac{T(W,1)}{T(W,P)} = \frac{1}{f + (1-f)/P}$$
$$\therefore S_P \rightarrow \frac{1}{f} \quad (P \rightarrow \infty)$$

Isogranular (Weak) Scaling

Solve a larger problem within the same time
duration using more processors $E_P = \frac{T(w,1)}{T(Pw,P)} \leq 1$

$W_P = Pw$ (weak scaling)

$w = \text{constant workload per processor (granularity)}$

• **Speedup:** $S_P = \frac{S(P \cdot w, P)}{S(w, 1)} = \frac{P \cdot w / T(P \cdot w, P)}{w / T(w, 1)} = \frac{P \cdot T(w, 1)}{T(P \cdot w, P)}$

• **Efficiency:** $E_P = \frac{S_P}{P} = \frac{T(w, 1)}{T(P \cdot w, P)}$

$$S_P = \frac{S(W_P, P)}{S(W_1, 1)} = \frac{W_P T(W_1, 1)}{W_1 T(W_P, P)}$$

$$E_P = \frac{S_P}{P} = \frac{W_P T(W_1, 1)}{P W_1 T(W_P, P)}$$

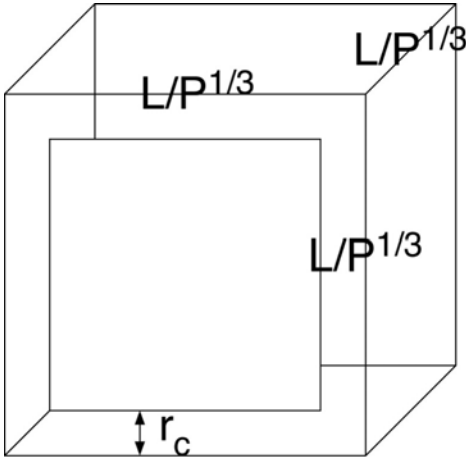
Analysis of Parallel MD

- **Parallel execution time:**
Workload $\propto$ Number of atoms, N (linked-list cell algorithm)

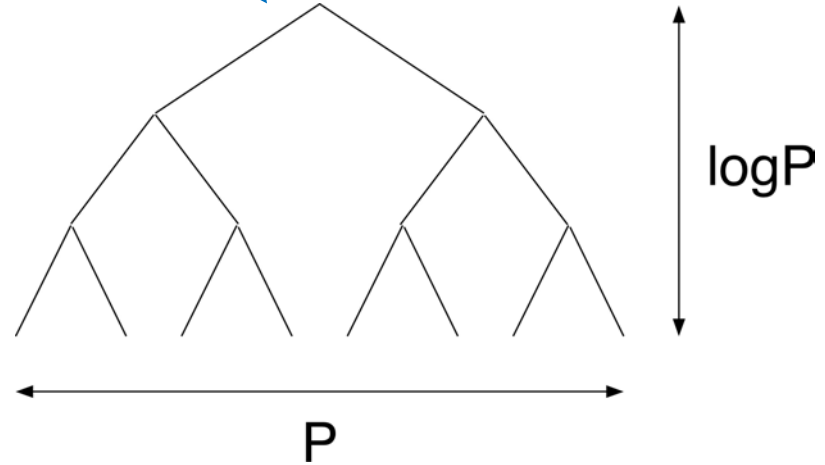
$$\begin{aligned}
 T(N, P) &= T_{\text{comp}}(N, P) + T_{\text{comm}}(N, P) + T_{\text{global}}(P) \\
 &= a \frac{N}{P} + b \left(\frac{N}{P} \right)^{2/3} + c \log P
 \end{aligned}$$

MPI_Allreduce()

Surface-to-volume ratio



Divide & conquer tree-height



cached volume
 $\frac{L^2}{P^{2/3}} r_c$ × atom density $\tilde{\rho}$
 $= 6r_c \frac{N^{2/3}/\rho^{2/3}}{P^{2/3}} \rho$
 $= 6r_c \rho^{1/3} \left(\frac{N}{P} \right)^{2/3}$

$\left(\because \frac{N}{L^3} = \rho \Rightarrow L^2 = \frac{N^{2/3}}{\rho^{2/3}} \right)$

Fixed Problem-Size (Strong) Scaling

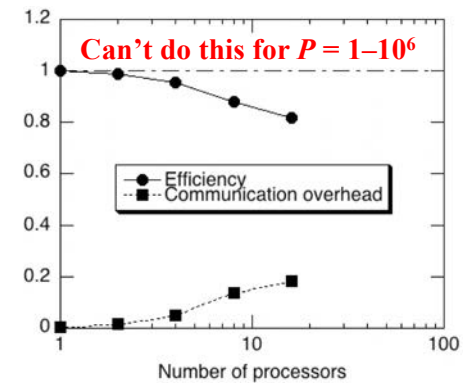
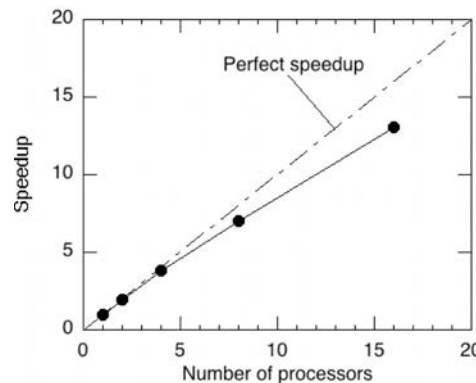
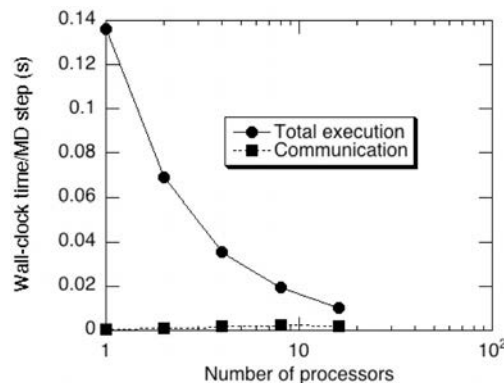
- **Speedup:**

$$S_P = \frac{T(N,1)}{T(N,P)} = \frac{aN}{aN/P + b(N/P)^{2/3} + c \log P}$$

$$= \frac{1}{1 + \frac{b}{a} \left(\frac{P}{N}\right)^{1/3} + \frac{c}{a} \frac{P \log P}{N}}$$

- **Efficiency:**

$$E_P = \frac{S_P}{P} = \frac{1}{1 + \frac{b}{a} \left(\frac{P}{N}\right)^{1/3} + \frac{c}{a} \frac{P \log P}{N}}$$

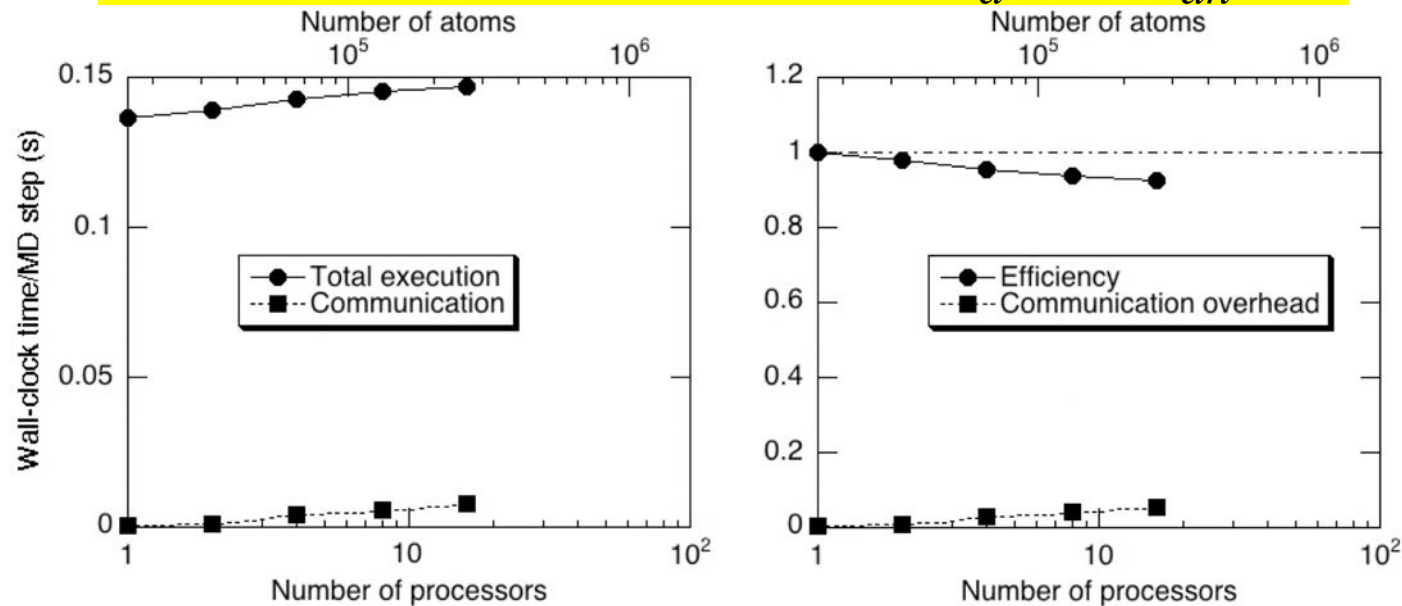


pmd.c: $N = 16,384$, on HPC (predecessor of CARC)

Isogranular (Weak) Scaling of Parallel MD

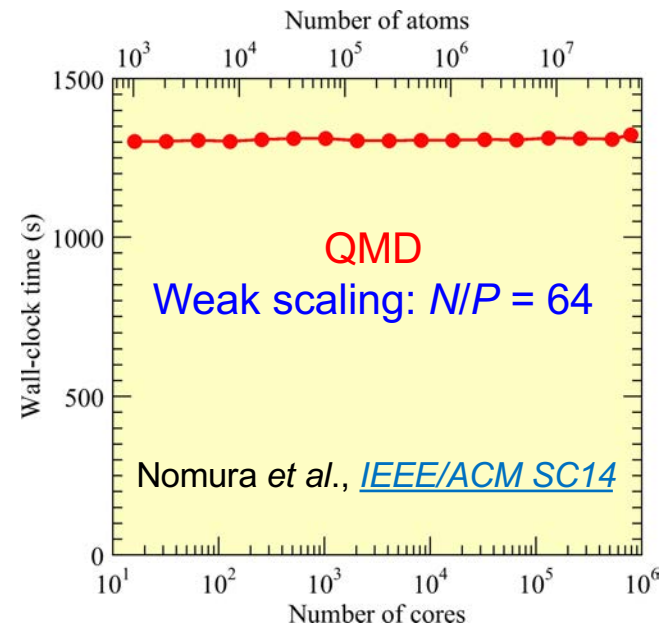
- $n = N/P = \text{constant}$: doable for arbitrarily large P
- Efficiency:

$$E_P = \frac{T(n,1)}{T(nP,P)} = \frac{an}{an + bn^{2/3} + c \log P} = \frac{1}{1 + \frac{b}{a}n^{-1/3} + \frac{c}{an} \log P}$$



pmd.c: $N/P = 16,384$, on HPC (predecessor of CARC)

High-End Parallel MD



- 4.9 trillion-atom space-time multiresolution MD (MRMD) of SiO_2
 - 8.5 billion-atom fast reactive force-field (F-ReaxFF) RMD of RDX
 - 39.8 trillion grid points (50.3 million-atom) DC-DFT QMD of SiC
- parallel efficiency 0.984 on 786,432 Blue Gene/Q cores

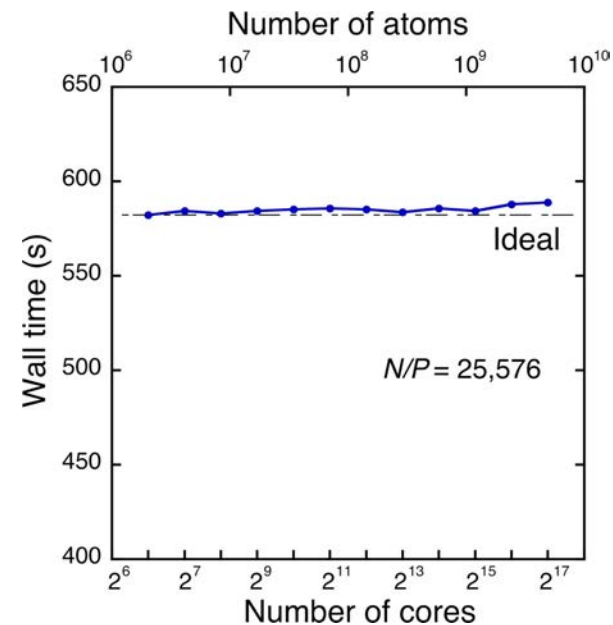
MD (molecular dynamics): MRMD

RMD (reactive molecular dynamics): F-ReaxFF

QMD (quantum molecular dynamics): DC-DFT

Portable Parallel Efficiency

- Weak-scaling parallel efficiency of 0.989 for a new generation of reactive molecular dynamics (RMD) on 131,072 Intel Knights Landing cores on Theta supercomputer at Argonne National Laboratory



K. Liu *et al.*, [Shift-collapse acceleration of generalized polarizable reactive molecular dynamics for machine learning-assisted computational synthesis of layered materials](#),
Proc. ScalA18 (IEEE, '18)

Quantum MD@Scale

Quantum dynamics at scale: ultrafast control of emergent functional materials

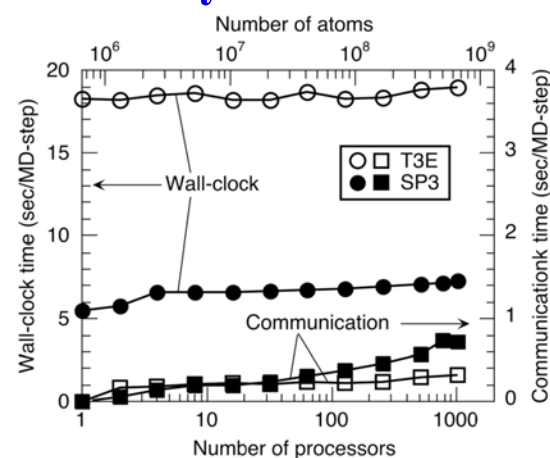
S. C. Tiwari, P. Sakdhnagool, R. K. Kalia, A. Krishnamoorthy, M. Kunaseth, A. Nakano, K. Nomura, P. Rajak, F. Shimojo, Y. Luo & P. Vashishta

Best Paper in *ACM HPC Asia 2020*



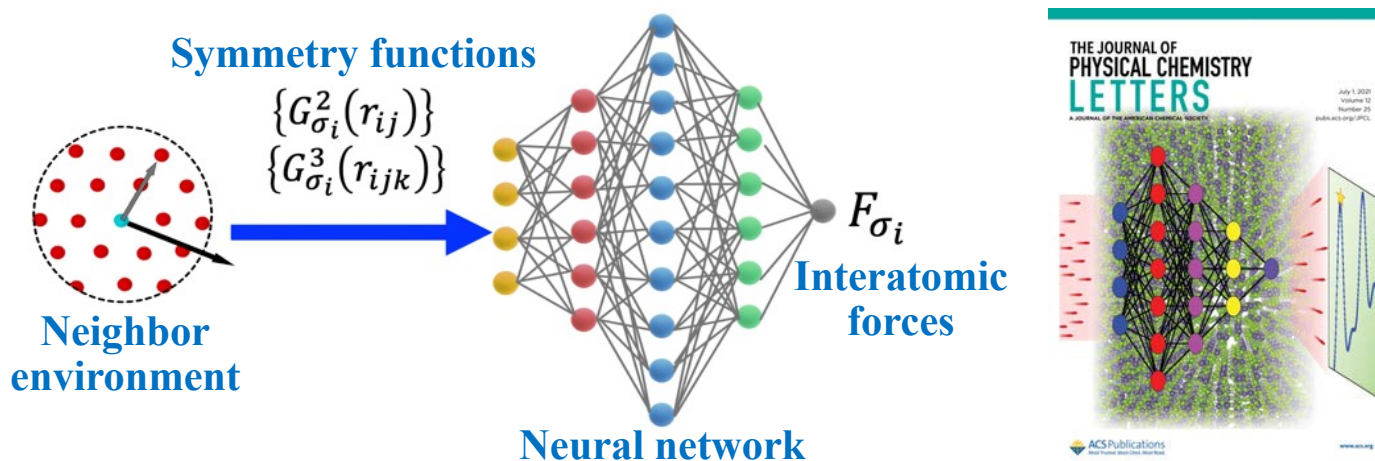
[Scalable atomistic simulation algorithms for materials research](#), A. Nakano *et al.*,
Best Paper, *IEEE/ACM Supercomputing 2001*, SC01

19 years since



Neural MD@Scale

- Neural-network quantum molecular dynamics (NNQMD) could revolutionize atomistic modeling of materials, providing quantum-mechanical accuracy at a fraction of computational cost [*Phys. Rev. Lett.* **126**, 216403 ('21); *J. Phys. Chem. Lett.* **12**, 6020 ('21); *Nature Commun.* **15**, 3911 ('24)]



[Neural network molecular dynamics at scale](#) & [Ex-NNQMD: extreme-scale neural network quantum molecular dynamics](#),

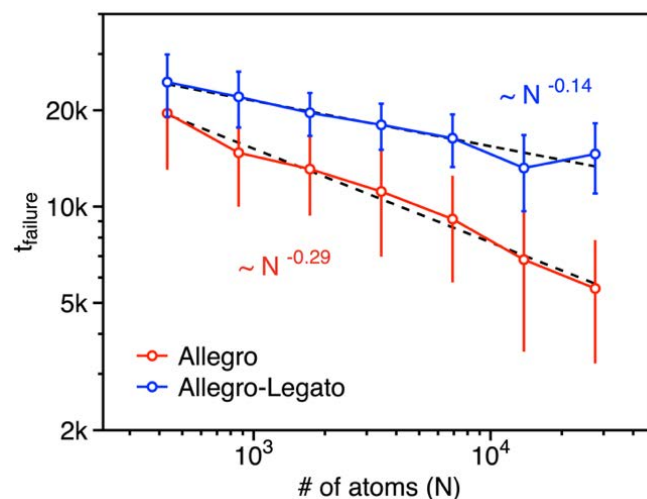
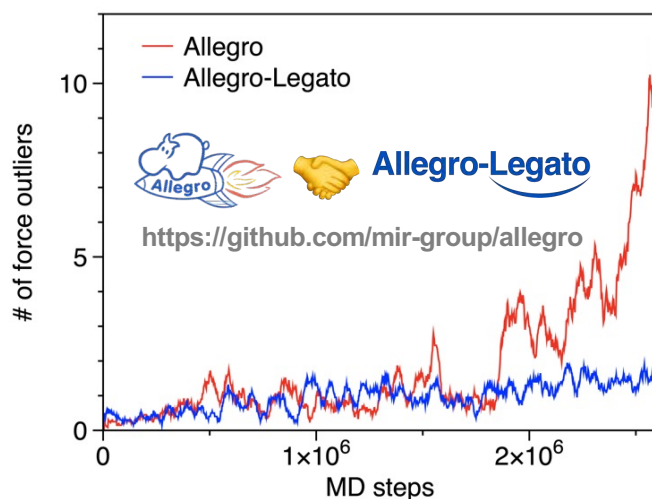
P. Rajak et al., *IEEE IPDPS ScaDL 20 & 21*

See also [Pushing the limit of molecular dynamics with ab initio accuracy to 100 million atoms with machine learning](#)

W. Jia et al., *ACM/IEEE Supercomputing, SC20*

Fast, Robust & Scalable: Allegro-Legato

- **Allegro (fast) NNQMD: State-of-the-art accuracy & speed** founded on group-theoretical equivariance & local descriptors Musaelian *et al.*, *Nat. Commun.* **14**, 579 ('23)
 - **Fidelity-scaling problem: On massively parallel computers, growing number of unphysical (adversarial) force predictions prohibits simulations involving larger numbers of atoms for longer times**
 - **Allegro-Legato (fast and “smooth”): Sharpness aware minimization (SAM)** enhances the **robustness** of Allegro through improved smoothness of loss landscape
- $\mathbf{w}_* = \operatorname{argmin}_{\mathbf{w}} [L(\mathbf{w}) + \max_{\|\epsilon\|_2 \leq \rho} \{L(\mathbf{w} + \epsilon) - L(\mathbf{w})\}]$ (L : loss; $\mathbf{w}$: model parameters)
- **Elongated time-to-failure scaling, $t_{\text{failure}} = O(N^{-\beta})$, without sacrificing accuracy or speed, thereby achieving spectroscopically stable long-time Hamiltonian trajectory**

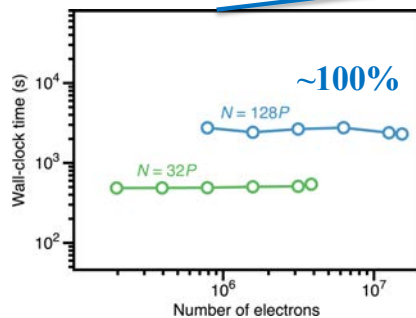
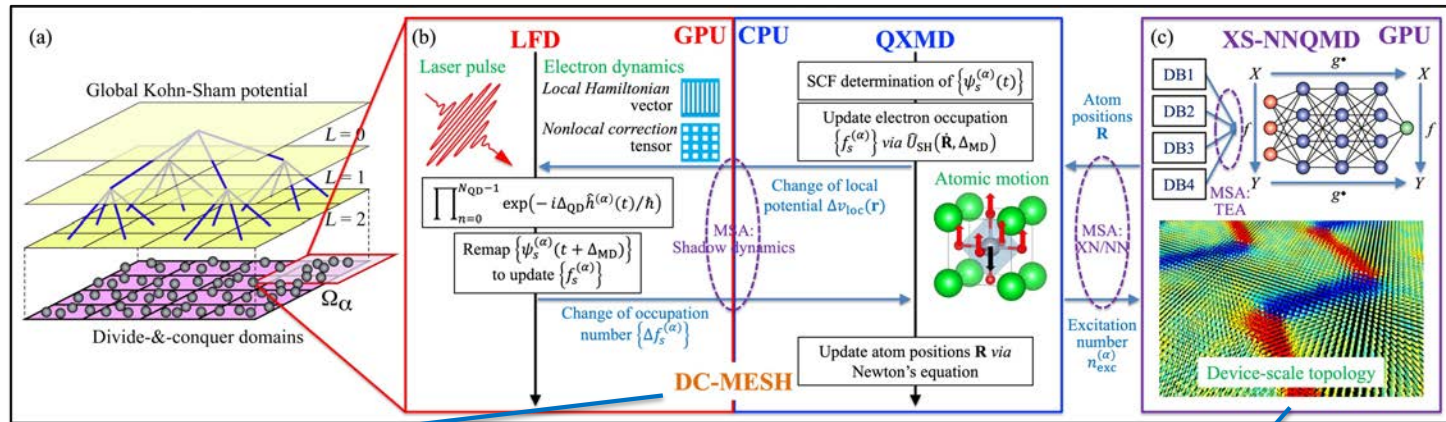


H. Ibayashi *et al.*, *ISC23—LNCS* **13948**, 223 ('23); *arXiv*: 2303.08169

Multi-Exaflop/s Light-Matter Dynamics

- Achieving multi-Exaflop/s:** On 60,000 GPUs of Aurora: **1.87 Exaflop/s**; $152\times$ & $3,780\times$ improvements of time-to-solution over state-of-the-art for 15.4M-electron DC-MESH NAQMD & 1.2T-atom excited XS-NNQMD; **nearly perfect parallel efficiency**

Exaflop/s = 10^{18} mathematical operations per second



DC-MESH NAQMD



**T. M. Razakh et al.,
2025 Gordon Bell prize finalist**

XS-NNQMD

